

# Who granted the days, who took them, and who never had them?

## An analyst case study — Vantage Wholesale Supply receivables

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**Reporting date:** 2025-12-31 · **Tooling:** T-SQL (SQL Server), Power Query, Excel

**Data disclosure.** Vantage Wholesale Supply is a fictional industrial parts distributor. Every figure in this study is synthetic, generated by the scripts in `sql/`, and reproducible from them. No confidential data is used and no claim is made about any production system.

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### 1. The situation

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Vantage Wholesale Supply sells bearings, fasteners, fluid-power components and MRO consumables on credit to around 400 trade accounts. Receivables run at roughly \$10.8M against credit sales of about \$176,000 a day.

Over the last seven quarters, Days Sales Outstanding has climbed from **55.85 to 63.07 days**. The finance director reads that as collections going slack. The collections manager reads it as Sales having sold longer terms and nobody having re-baselined the target. Both are plausible, both are self-serving, and the argument has been running on assertion because nobody has split the number.

Meanwhile six collectors work a 2,300-line ageing report from the top down by balance.

### The question

**Of the days DSO has risen, how many did Vantage grant itself through the terms it sold, and how many are customers taking beyond those terms — and given that split, who should each collector call first?**

The answer turned out to need a third category that neither side had proposed.

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## 2. Approach

### 2.1 One definition of what is owed

Everything downstream reads from a single function, so the definition cannot drift into several:

```
OpenBalance = InvoiceAmount - cash applied - discounts taken - credit memos - write-offs
```

The word **applied** is doing real work there. Cash that has reached the bank but has not been matched to an invoice does *not* reduce a balance, because the invoice genuinely is still open — and the date that reduces it is the application date, never the receipt date. Cash banked on 28 December and matched on 4 January does not cure a December ageing bucket, and treating it as if it did is a well-worn way to flatter a month end.

### 2.2 Two DSOs, on purpose

| Method                                 | Reads          | Used for                                                                                                                                            |
|----------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Countback</b><br>(exhaustion)       | <b>63.07 d</b> | The headline. Consumes the balance against the actual months of sales that generated it, so a seasonal dip cannot manufacture a collections crisis. |
| <b>Classic</b> (balance ÷ daily sales) | <b>61.59 d</b> | The bridge, and any external comparison. It is what a lender or a benchmark quotes.                                                                 |

Quoting a countback figure to someone benchmarking on classic starts an argument you will lose, so both are published and the gap is itself a reported number.

### 2.3 The bridge, built to be exact

The decomposition is deliberately built on **classic** DSO, because classic decomposes exactly and countback does not.  $DSO = TotalAR \div SalesPerDay$ , and total AR partitions with no overlap and nothing left over:

| Component                            | Days         | Cash                   | Owner                |
|--------------------------------------|--------------|------------------------|----------------------|
| <b>Granted</b> — not yet due         | 45.97        | \$8,095,793.78         | Sales / terms policy |
| <b>Taken</b> — disputed and past due | 0.88         | \$154,864.71           | Billing / sales      |
| <b>Taken</b> — past due, undisputed  | 14.74        | \$2,596,423.00         | Collections          |
| <b>Total (classic DSO)</b>           | <b>61.59</b> | <b>\$10,847,081.49</b> |                      |

No residual bar, nothing to plug. The identity is asserted on every test run (UAT-04).

**Billing lag is deliberately not in this table.** DSO counts from the invoice date, so the days between despatching goods and issuing the invoice happen before the clock starts. Folding them in would break the identity. They *extend* it instead:

CashCycle = BillingLag + DSO\_classic = 1.78 + 61.59 = 63.37 days

That is the honest figure — despatch to cash — and it is larger than the number anyone at Vantage is currently reporting.

2.4 Every number computed twice

Each headline figure is implemented independently in T-SQL and again in Excel formulas, then reconciled on the workbook's Validation sheet, in the open. All **27** checks agree. The one worth pausing on: SQL computes countback DSO with a windowed running sum and a TOP 1 ; Excel computes it with a twelve-row helper block, AGGREGATE , MATCH and INDEX . They agree to **0.0011 days**.

3. What the data says

3.1 The split — and the third category

Over the comparable window (2024-06 to 2025-12; the ledger opens in January 2024 and cannot show a representative ageing profile before late April):

|                                                 | 2024-06 | 2025-12 | Change |
|-------------------------------------------------|---------|---------|--------|
| DSO (countback)                                 | 55.85   | 63.07   | +7.22  |
| of which <b>granted</b> (Best Possible DSO)     | 44.33   | 46.27   | +1.94  |
| of which <b>taken</b> (Average Days Delinquent) | 11.52   | 16.80   | +5.28  |

So the collections manager is more right than the finance director: roughly a quarter of the rise was granted, three quarters is customers paying later.

**Except that a large part of the "taken" is not customers at all.**

|                           | 2024-06 | 2025-12             |
|---------------------------|---------|---------------------|
| Unapplied cash as % of AR | 1.79%   | <b>3.87%</b>        |
| Unapplied cash            | —       | <b>\$419,392.13</b> |

At \$176,118 of credit sales per day, \$419,392 of unmatched cash is **2.38 days of DSO that no customer is responsible for**. It is Vantage's money, in Vantage's bank account, making Vantage's invoices look unpaid.

**The honest decomposition is therefore three ways, not two:** of the 16.80 days sitting in "taken", **2.38 are self-inflicted** and 14.42 are genuine customer lateness.

That third category could not even be expressed until cash receipts and cash applications were modelled as separate events. Under the original single-table design, a receipt and its application were the same row, so "money arrived but nobody matched it" had no representation — and the 2.38 days were silently charged to customers.

### 3.2 The granted quarter was a decision, not an accident

Weighted average terms rose 2.28 days between the second halves of 2024 and 2025. A shift-share decomposition separates *who bought* from *what they were sold*:

| Effect                                       | Days         | Reading                                    |
|----------------------------------------------|--------------|--------------------------------------------|
| Mix — same terms, different customers buying | −0.33        | Slightly offsetting                        |
| <b>Rate — same customers, longer terms</b>   | <b>+1.71</b> | <b>A policy change</b>                     |
| Interaction                                  | +0.90        | Re-papered customers also grew their share |
| <b>Total ΔWAT</b>                            | <b>+2.28</b> |                                            |

All 400 customers traded in both periods, so this is not composition. The same accounts were moved onto longer terms, and the accounts that were moved bought more. That is a commercial decision someone made; it should be priced and capped rather than discovered in the DSO line.

### 3.3 The DSO target cannot be hit — arithmetically

|            | Days  |
|------------|-------|
| DSO target | 45.00 |

| Days                                          |              |
|-----------------------------------------------|--------------|
| Weighted average terms actually sold          | <b>46.68</b> |
| <b>Collection allowance the target leaves</b> | <b>-1.68</b> |

If every customer paid **exactly on the due date**, DSO would settle near 46.7 and the 45-day target would still be missed. The target sits below the floor the terms mix permits.

A target nobody can reach does not motivate a collections team. It teaches them the scorecard is theatre, which costs more than the target was ever worth.

### 3.4 Three metrics are Green or Amber while a cohort inside them has failed

This is the pattern worth taking away from the whole engagement.

| Metric             | Portfolio | Status       | The cohort underneath                                              |
|--------------------|-----------|--------------|--------------------------------------------------------------------|
| Promise kept rate  | 86.29%    | <b>Green</b> | High-risk tier: <b>53.87%</b> by value, 40.29% by count            |
| Billing lag        | 1.78 d    | <b>Green</b> | Southeast DC: <b>3.22 d</b> , drifting 1.56 → 4.78 through 2025    |
| Credit utilisation | 17.70%    | <b>Green</b> | One account over limit at <b>106%</b> ; three above the 80% target |

Every one of these aggregates is arithmetically correct and every one is operationally useless on its own. A portfolio average cannot breach until the whole book does; by then the cohort that caused it has been failing for a year. Each of these three now ships with its cohort cut attached to the same report, because a number that can only be read one way is a number that will be read the wrong way.

The promise figure is the sharpest: **244 broken promises worth \$625,642**, almost entirely in one risk tier, sitting underneath a portfolio figure that clears its target.

### 3.5 Collections effectiveness is flattered by write-offs

The textbook Collection Effectiveness Index measures whether AR went down — and AR goes down identically whether a customer wired the money or a controller wrote the balance off. That makes the standard metric improvable by giving up.

|                 | 2024-06 | 2025-12 |
|-----------------|---------|---------|
| CEI (book)      | 71.80%  | 66.91%  |
| CEI (cash only) | 71.05%  | 65.26%  |

|                       | 2024-06  | 2025-12  |
|-----------------------|----------|----------|
| Paper collections gap | 0.75 pts | 1.65 pts |

Write-offs were zero until April 2025 and total **\$702,389** since. Both figures are below the 75% warning threshold; the cash figure is further below, and it is the honest one.

### 3.6 The receivables balance is mis-stated by \$371,040

The data-quality gate **fails**, and deliberately is not suppressed:

| Defect                               | Count | Value        |
|--------------------------------------|-------|--------------|
| Duplicate cash receipts              | 96    | \$371,039.99 |
| Over-applied invoices (their effect) | 101   | \$371,039.99 |
| Unapplied cash older than 90 days    | 65    | \$265,487.11 |
| Receipts dated before their invoice  | 30    | \$100,496.95 |
| Invoices issued before despatch      | 20    | \$76,841.74  |
| Due date inconsistent with terms     | 52    | \$12,727.40  |

The duplicates are the material one: **3.421% of open AR**, against a 1.000% tolerance. Note that `DUPLICATE_RECEIPT` and `OVER_APPLIED_CASH` report the *same* \$371,039.99 — one is the cause and the other its effect on the ledger. Only the effect counts toward exposure, or the same money would be charged twice.

## 4. The operational control

A ranked list of every account with a past-due balance is a ledger with a sort order. 272 accounts is two thirds of the customer base, and six collectors cannot call that in a day.

The queue ranks on **collectable exposure** — ageing-weighted past-due dollars, less 75% of the ageing-weighted *disputed* dollars, less cash already banked against the account — and then bounds itself to what the team can work:

| Action                | Accounts | Past due  | Collectable exposure | On today's list |
|-----------------------|----------|-----------|----------------------|-----------------|
| <code>ESCALATE</code> | 19       | \$758,532 | \$359,632            | <b>19</b>       |

| Action            | Accounts   | Past due           | Collectable exposure | On today's list |
|-------------------|------------|--------------------|----------------------|-----------------|
| COLLECTION_CALL   | 26         | \$476,443          | \$155,712            | 15              |
| STANDARD_DUNNING  | 100        | \$634,679          | \$144,147            | 21              |
| FINAL_NOTICE      | 19         | \$356,211          | \$80,901             | 5               |
| RESOLVE_DISPUTE   | 45         | \$195,694          | \$36,407             | 5               |
| COURTESY_REMINDER | 12         | \$26,036           | \$6,509              | 0               |
| APPLY_CASH        | 51         | \$303,692          | \$5,011              | 0               |
| <b>Total</b>      | <b>272</b> | <b>\$2,751,288</b> | <b>\$788,318</b>     | <b>65</b>       |

**65 accounts — 10 to 12 per collector — carrying 81.3% of the collectable exposure.**

Four design decisions are worth naming:

- **APPLY\_CASH outranks everything.** 51 accounts have cash already banked against them

covering at least half their past-due balance. Ringing a customer who has already paid is the single most damaging call a collections team can make, and it is not a collections job at all — it is a cash application job. These accounts are routed out of the call list entirely, and **UAT-20 asserts that none of them can reach a collector's worklist.**

- **Disputes are discounted, not ranked.** Chasing cash on a disputed invoice does not work; that balance needs the dispute resolving, which is a different team's job.

- **Two broken promises in 90 days escalates,** regardless of balance. It is the earliest reliable distress signal in the data and it costs nothing to collect.

- **Credit hold is a flag, not an action.** Holding orders sits alongside a collections action rather than replacing it. As a competing branch it was also unreachable — every over-limit account already qualified for escalation — so the control silently did nothing.

## The second worklist

Collections gets a call list. Cash application gets its own, ranked oldest-first:

**97 accounts hold \$419,392 of unmatched cash. 57 receipts have no application against them at all. The oldest has been sitting for 679 days. \$409,640 of past-due balance would be wrongly chased if nobody applies it first.**

This is the cheapest money in the entire analysis: no negotiation, no concession, no relationship cost. It is already in the bank.

## 5. Recommendations

| # | Recommendation                                                            | Owner               | Basis                                                                                 | Value                                                                                          |
|---|---------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 | <b>Apply the cash before making a single call</b>                         | Cash application    | 97 accounts hold \$419,392 unmatched; 57 receipts wholly unapplied; oldest 679 days   | <b>2.38 days of DSO</b> and <b>\$409,640</b> of wrongly-chaseable balance, at no customer cost |
| 2 | <b>Fix duplicate receipt posting</b>                                      | Controller          | 96 duplicate bankings mis-state AR by 3.421%                                          | <b>\$371,040</b> of AR correctly stated                                                        |
| 3 | <b>Reset the DSO target to terms + allowance, about 55 days</b>           | CFO                 | Target 45 sits 1.68 days below the 46.68-day terms floor                              | Restores a scorecard the team can act on                                                       |
| 4 | <b>Work the bounded queue</b> , starting with the 19 escalations          | Collections manager | 65 accounts carry 81.3% of collectable exposure                                       | Focuses 6 collectors on <b>\$640,540</b> of the \$788,318 at stake                             |
| 5 | <b>Report promise kept-rate by risk tier, never as a portfolio figure</b> | Collections manager | 86.29% overall conceals 53.87% in the High tier, 244 broken promises worth \$625,642  | Turns a Green light into an early-warning system                                               |
| 6 | <b>Investigate Southeast invoice posting</b>                              | Operations          | 3.22 days against a 1.50 baseline elsewhere, drifting all year                        | <b>\$50,477</b> of cash, recoverable by changing a batch schedule                              |
| 7 | <b>Price and cap the extended-terms programme</b> at the Q1 review        | Sales director      | Shift-share isolates a deliberate +1.71-day rate effect across the same 400 customers | Makes ~\$301k of working capital a priced decision rather than a drift                         |
| 8 | <b>Measure collections on CEI (cash), not CEI (book)</b>                  | CFO                 | \$702,389 of write-offs currently score as collections                                | Removes an incentive to improve the metric by giving up                                        |

Recommendations 1 and 2 are the same root cause — cash application is the broken function — and together they account for **\$790,432** of mis-stated or unusable receivables. Neither needs a customer conversation.

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## 6. What went wrong during the build, and what it cost

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### 6.1 A false finding that survived every balance check

The generator paid 60% of a partial invoice on the first instalment and emitted the second only when the invoice was *not* a discount-taker — the amount rule and the row filter disagreed about the same condition. 208 invoices kept a 40% residual that could never clear.

Because `2/10NET30` is the only terms code carrying a discount, the whole defect landed on one cohort and presented as a clean commercial insight:

*"47.4% of open discount-terms invoices are more than 90 days past due, against 2.3% on NET30."*

Every row was internally consistent. The settlement identity held to the cent. The ageing buckets summed correctly. No balance-level check could see it. It surfaced only when open AR was cut by **ageing and terms together** and the result was too extreme to be plausible behaviour.

**The lesson:** a defect that is internally consistent will pass every internal check. Cross-cuts that no single check covers are where they surface — and an implausible result deserves investigation before it deserves a slide.

### 6.2 A dashboard that reported success while broken

The Excel build script completed without error and printed "Saved". Opening the file showed `#REF!` through the entire Priority Queue: Excel does not defer a reference to a sheet that does not exist yet — it rewrites the reference and moves on, and the Calc sheet had been written before the sheets it referenced were created.

A second fault was worse because it was quieter: the ageing matrix's dimension lists were written over the countback block's "covers the balance?" flags, so `MATCH` skipped the overwritten rows and returned Best Possible DSO of **42.75** instead of **46.19** — a plausible number, not an error.

Both were found by a separate script that opens the saved workbook, forces a recalculation and reads the values back. **A build that finishes without throwing has proved it ran, not that it produced a working file.**

### 6.3 A published figure that disagreed with its own components

`AvgDaysDelinquent` is `DSO - BPDSO`. All three were rounded to two decimals independently, so the page published 63.07, 46.27 and 16.81 — and a reader doing the subtraction got 16.80. Caught

by UAT-11.

The fix is a rule now applied in both implementations: **a derived figure is computed from the components as published, not from their unrounded intermediates**. The same class of defect had already appeared once, where `fn_DSO` divided before multiplying while `fn_DS0Bridge` multiplied before dividing, and the two published a classic DSO that differed by a cent.

## 6.4 A test whose premise my own feature invalidated

UAT-02 asserted as-of integrity by subtracting all cash applied after 30 June from the December figure. Once applications could legitimately fall *after* the reporting date — the December-banked, January-applied case the new model exists to represent — that subtraction removed \$165,415.55 that had never been in the December total. **The test failed; the data was right.**

It was replaced with a direct recomputation rather than a subtraction. Worth recording because the instinct on a red test is to suspect the code, and here the code was the only correct part.

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## 7. What I would do differently

- **Model cash receipt and cash application separately from the start.** This was originally deferred as a limitation, and closing it changed a headline conclusion: 2.38 days of DSO moved out of "customers are paying late" and into "we have not applied our own cash". A modelling shortcut that cannot represent a category will silently attribute that category to something else.
  - **Cut the data by pairs of dimensions during profiling.** The instalment defect was invisible to single-dimension checks and obvious the moment ageing met terms. Every cohort finding in §3.4 came from the same habit, applied deliberately rather than accidentally.
  - **Write the workbook validator before the workbook.** It found two defects in the first file it opened. Had it existed first, the build would never have reported a success it had not earned.
  - **Treat rounding as a published contract, not a formatting choice.** Two of the defects above were rounding inconsistencies that produced figures which did not add up on the page.
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## 8. How to verify any figure in this study

```
sqlcmd -S <server> -E -d master -i sql/01_create_schema.sql
sqlcmd -S <server> -E -d master -i sql/02_generate_synthetic_data.sql
sqlcmd -S <server> -E -d VantageAR -i sql/03_core_views.sql
sqlcmd -S <server> -E -d VantageAR -i sql/04_data_quality_checks.sql
sqlcmd -S <server> -E -d VantageAR -i sql/05_kpi_views.sql
sqlcmd -S <server> -E -d VantageAR -i sql/06_dso_bridge.sql
sqlcmd -S <server> -E -d VantageAR -i sql/07_stored_procedures.sql
sqlcmd -S <server> -E -d VantageAR -i sql/08_uat_test_cases.sql # 25 cases

EXEC dbo.usp_ARScorecard @AsOf = '2025-12-31';
EXEC dbo.usp_DSABridgeReport @AsOf = '2025-12-31';
EXEC dbo.usp_PriorityQueue @AsOf = '2025-12-31', @TodaysWorklistOnly = 1;
EXEC dbo.usp_CashApplicationWorklist @AsOf = '2025-12-31';
EXEC dbo.usp_PromiseReport @AsOf = '2025-12-31', @GroupBy = 'RiskTier';
EXEC dbo.usp_BillingLagReport @AsOf = '2025-12-31';
EXEC dbo.usp_RunDataQualityChecks;

powershell erp_extracts/Export_Extracts.ps1
powershell excel/Build_Workbook.ps1
powershell excel/Validate_Workbook.ps1 # 27 checks
```

The generator is deterministic — hash-keyed rather than `RAND()` — so a rebuild reproduces the dataset byte for byte, and UAT-16 asserts the fingerprints.

## Appendix — headline figures at 2025-12-31

| Metric                          | Value        | Target | Warning | Status |
|---------------------------------|--------------|--------|---------|--------|
| DSO (countback)                 | 63.07 d      | 45.00  | 55.00   | Red    |
| Average days delinquent         | 16.80 d      | 8.00   | 15.00   | Red    |
| Collection effectiveness (book) | 66.91%       | 85.00  | 75.00   | Red    |
| Collection effectiveness (cash) | 65.26%       | —      | —       | —      |
| Past due share of AR            | 25.36%       | 15.00  | 25.00   | Red    |
| 90+ share of AR                 | 3.35%        | 3.00   | 6.00    | Amber  |
| <b>Unapplied cash % of AR</b>   | <b>3.87%</b> | 0.50   | 1.50    | Red    |
| Billing lag (days)              | 1.78 d       | 2.00   | 4.00    | Green* |

| Metric                         | Value   | Target | Warning | Status |
|--------------------------------|---------|--------|---------|--------|
| Promise kept rate              | 86.29%  | 80.00  | 65.00   | Green* |
| Credit utilisation (portfolio) | 17.70%  | 80.00  | 100.00  | Green* |
| Mean open dispute age          | 28.13 d | 15.00  | 30.00   | Amber  |

\\* Green at portfolio level only. Southeast billing lag is 3.22 days; the High-risk promise kept rate is 53.87%; three accounts are over their credit limit, one at 106%. See §3.4.

| Ageing bucket | Invoices     | Balance                | Share       |
|---------------|--------------|------------------------|-------------|
| Current       | 1,672        | \$8,095,793.78         | 74.6%       |
| 1–30 days     | 414          | \$1,839,519.41         | 17.0%       |
| 31–60 days    | 101          | \$405,938.13           | 3.7%        |
| 61–90 days    | 34           | \$142,966.39           | 1.3%        |
| 90+ days      | 125          | \$362,863.78           | 3.3%        |
| <b>Total</b>  | <b>2,346</b> | <b>\$10,847,081.49</b> | <b>100%</b> |

| Cash cycle                                    | Days         |
|-----------------------------------------------|--------------|
| Billing lag (despatch → invoice)              | 1.78         |
| Classic DSO (invoice → cash applied)          | 61.59        |
| <b>True cash cycle (despatch → cash)</b>      | <b>63.37</b> |
| of which unapplied cash Vantage already holds | 2.38         |